Bangladesh University of Business and Technology
Department of Computer Science and Engineering

Course Title: Database Systems Course Code: CSE 207

Time: 30 Minutes **Set: A** **Marks: 15**

[CLO1] **1.** A university is developing an online course registration system that will be accessed simultaneously by thousands of students through web browsers. **Apply the concepts of two-tier and three-tier database application architectures** to design the system. Identify the components involved in each architecture and show how the client, application server, and database server would communicate. Based on the given scenario, select the more appropriate architecture and justify your choice. [5]

[CLO1] **2.** **Consider the following relational schema:** **[5]**

Course (course_id, course_name, category, start_date, instructor_id)
Instructor (i_id, i_name, total_courses)
Student (s_id, s_name, department)
Course_Student (course_id, s_id)

a) Create the Course and Course_Student tables with appropriate Primary Key and Foreign Key constraints.
b) Find the names of instructors who have taught courses belonging to the "Programming" category.
c) Write an SQL query to find the number of students enrolled in each course. Display the course_id, course_name, and the total number of enrolled students.
d) Write an SQL query **to** find the categories that contain more than 2 courses. Display the category name and the number of courses in each category.
e) Write an SQL query to find the instructor(s) who have taught the highest number of courses**.** Display the instructor's name and the number of courses taught.

[CLO1] **2.** Using the relational schema given in Question 2, construct appropriate **relational algebra expressions** for the following: [5]

a) Relational Algebra Write a relational algebra expression to display the course name and instructor name for every course.
b) Write a relational algebra expression to find the names of instructors who teach courses in the "Database" category.
c) Write a relational algebra expression to retrieve the names of students, course names, and instructor names for all course enrollments.
d) Using extended relational algebra, write an expression to find the number of students enrolled in each course. Display the course_id, course_name, and number of students enrolled.
(e) Find the course categories that have more than 3 courses. Display each category and its number of courses.

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Time: 30 Minutes **Set: B** **Marks: 15**

[CLO1] **1.** Apply DDL and DML commands to perform database operations, and demonstrate the differences between DDL and DML with appropriate SQL examples. [5]

[CLO1] **2.** **Consider the following relational schema:** **[5]**

Doctor (doc_id, doc_name, specialization, joining_date, hospital_id)
Hospital (h_id, h_name, total_doctors)
Patient (p_id, p_name, city)
Doctor_Patient (doc_id, p_id)

a) Create the Doctor and Doctor_Patient tables with appropriate Primary Key and Foreign Key constraints.
b) Find the hospitals that have more than 5 doctors. Display the hospital name and the total number of doctors.
c) Find the number of patients treated by each doctor. Display the doctor's name and the number of patients. Include doctors who have not treated any patients.
d) Find the names of doctors who work in the hospital with the highest number of doctors**.** Display the doctor's name and specialization.
e) Find the total number of doctors working in each hospital. Display the hospital name and the number of doctors.

[CLO1] **2.** Using the relational schema given in Question 2, construct appropriate **relational algebra expressions** for the following: [5]

a) Find the names and specializations of all doctors whose specialization is 'Cardiology'.
b) Find the names of doctors and their specializations who have treated patients from 'Chittagong'.
c) Find the names of doctors who have treated patients from 'Dhaka'.
d) Find the hospital name and the number of doctors working in each hospital, and display only those hospitals where the number of doctors is greater than 5.
e) Find the names of doctors who work at the hospital having the maximum number of doctors.